

Results: Field Overview

Results: Field Overview I

The de-duplicated corpus for **Exoplanet Atmospheres** contains $N = 46$ records spanning 2010–2023 (13 years). Publication volume grows at a compound annual rate of 12.27% (mean year-over-year growth 63.3%, doubling time 1.4 years), peaking in 2023 with 9 records that year. The growth curve is a first-order descriptive summary of the retained corpus; it is not a field-wide estimate of research activity.

Results: Field Overview II

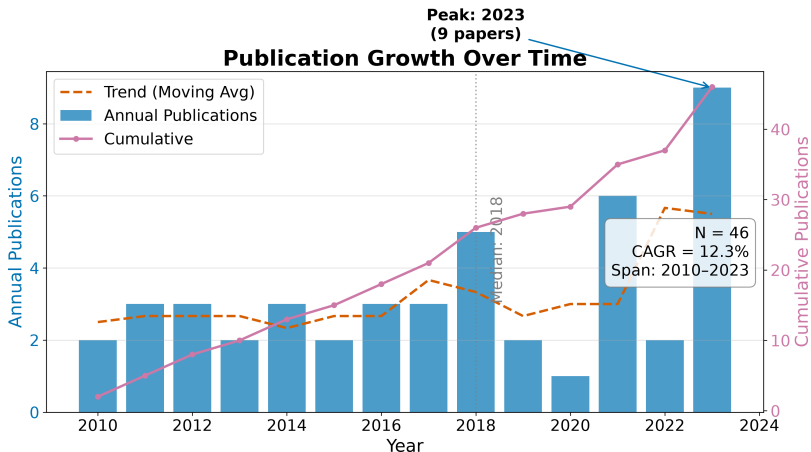


Figure 1: Publication growth curve for Exoplanet Atmospheres. Annual publication counts (bars) and cumulative total (line) show sustained growth from 2010 through 2023, peaking in 2023.

RQ1: Field Size and Growth I

The temporal analysis describes a retained literature slice spanning 13 years. The compound annual growth rate of 12.27% implies a corpus doubling time of approximately 1.4 years under the configured retrieval and indexing conditions. The peak year 2023 contains 9 retained records; that count may reflect both publication activity and delays or differences in source indexing, so it should not be interpreted as a causal trend without a live coverage audit.

Table 1. Top publication years.

Year	Publications
2010	2
2011	3
2012	3
2013	2
2014	3
2016	3
2017	3
2018	5
2021	6
2023	9

RQ2: Subfield Composition I

Records distribute across the 4 configured subfields as shown in Table 2, with **Observational Methods** the largest bucket at 50.0% of the classified corpus. The dominance of Observational Methods is a property of the configured taxonomy and retained corpus. It is not a domain prevalence estimate without a live, source-backed review and a documented coverage assessment.

Table 2. Subfield distribution.

Subfield	Papers	Share
Observational Methods	23	50.0%
Atmospheric Molecules	7	15.2%
Jwst Instruments	9	19.6%
Theoretical Modeling	7	15.2%

RQ2: Subfield Composition II

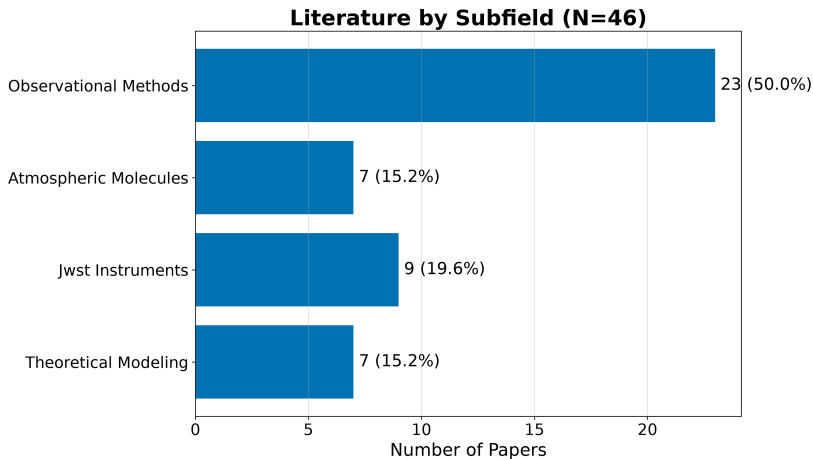
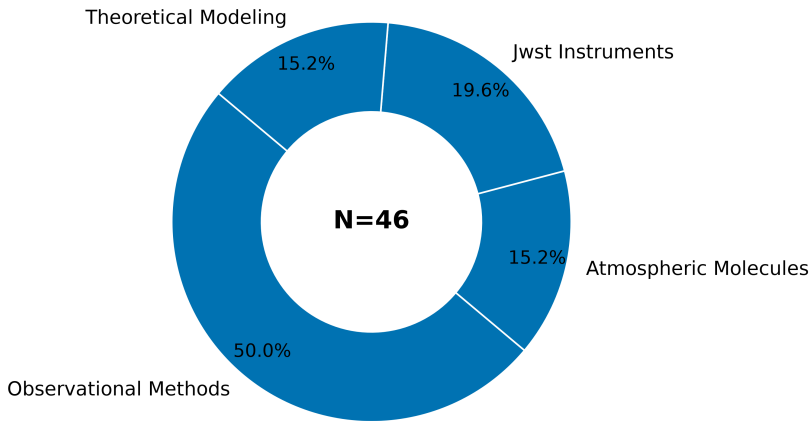


Figure 2: Field summary dashboard for Exoplanet Atmospheres. The dashboard combines corpus size, temporal range, subfield distribution, and key bibliometric indicators in a single overview panel.

RQ2: Subfield Composition III

Subfield Distribution



RQ2: Subfield Composition IV

Figure 3: Subfield distribution for Exoplanet Atmospheres. The 4-bucket taxonomy shows the relative weight of each configured sub-area, with Observational Methods dominant at 50.0%.

RQ2: Subfield Composition V

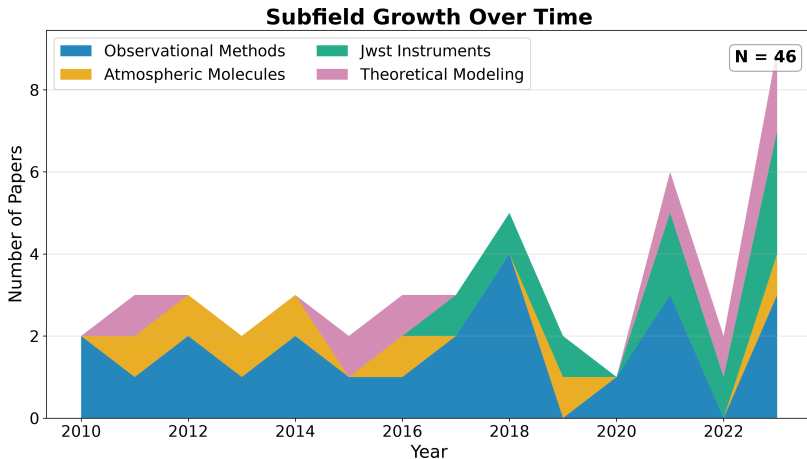


Figure 4: Subfield timeline for Exoplanet Atmospheres. Stacked annual publication counts by subfield show how each sub-area has evolved over time, revealing emerging and declining research threads.

Identifier and Full-Text Coverage I

The corpus has measurable identifier coverage: 46 of 46 records (100.0%) carry DOIs, supporting cross-engine de-duplication. OpenAlex IDs are present for 5 records. Abstract coverage stands at 100.0% (46 records), which limits the text analytics to that subset. Open-access status is confirmed for 43.5% of records, and 43.5% have a direct PDF link.

Descriptive Bibliometrics I

The corpus spans 104 unique authors across 46 papers, yielding a mean of 1.30 papers per author. Citation counts range from zero to 110 (mean 30.0, median 25.0), with a total of 1,381 citations across the corpus. The Gini coefficient of citation concentration is 0.474. This statistic describes concentration within the retained corpus and should not be generalized to citation behavior in the field.

Table 3. Citation count distribution.

Citations Papers	
0	0
1-9	12
10-49	24
50-99	9
100-499	1
500+	0

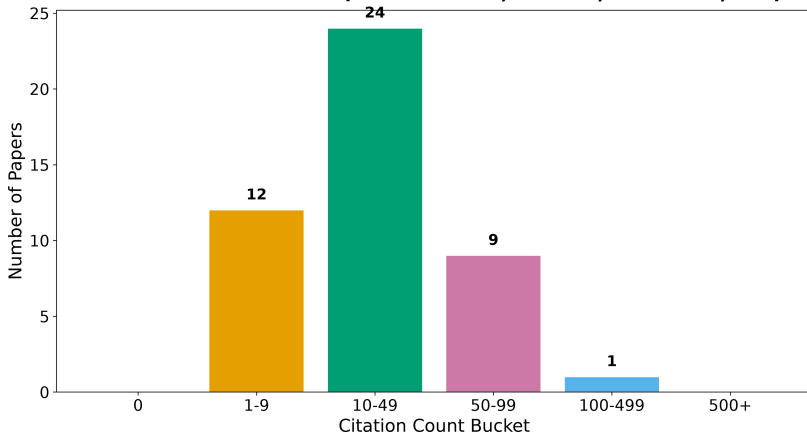
Citation Distribution (Gini = 0.474, N = 46, Total = 1,381)

Figure 5: Citation distribution for Exoplanet Atmospheres. The histogram shows the number of papers in each citation-count bucket, with the Gini coefficient annotated. The distribution is a descriptive property of the retained corpus.

Descriptive Bibliometrics III

Table 4. Top publication venues.

Venue	Papers
Astronomy and Astrophysics	8
Monthly Notices of the Royal Astronomical Society	7
Nature Astronomy	7
Publications of the Astronomical Society of the Pa	7
The Astrophysical Journal	5
Icarus	4
Research Notes of the AAS	4
The Astronomical Journal	4

Descriptive Bibliometrics IV

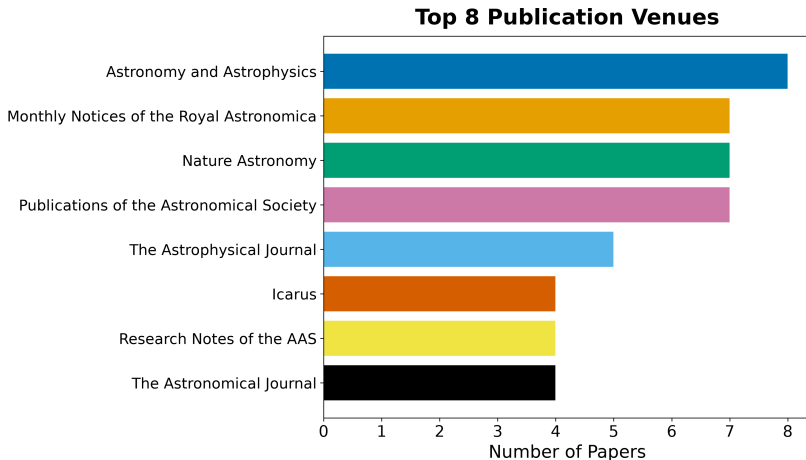


Figure 6: Top publication venues for Exoplanet Atmospheres. The horizontal bar chart shows the venues with the most retained records; it describes this corpus rather than the complete field.

Descriptive Bibliometrics V

Table 5. Top authors by publication count.

Rank	Author	Papers
1	D. Ito	3
2	G. Petrov	3
3	J. Ito	3
4	L. Carter	3
5	A. Fournier	2
6	A. Jensen	2
7	A. Singh	2
8	B. Kowalski	2
9	B. Owens	2
10	C. Esposito	2

Descriptive Bibliometrics VI

Top 20 Authors by Publication Count

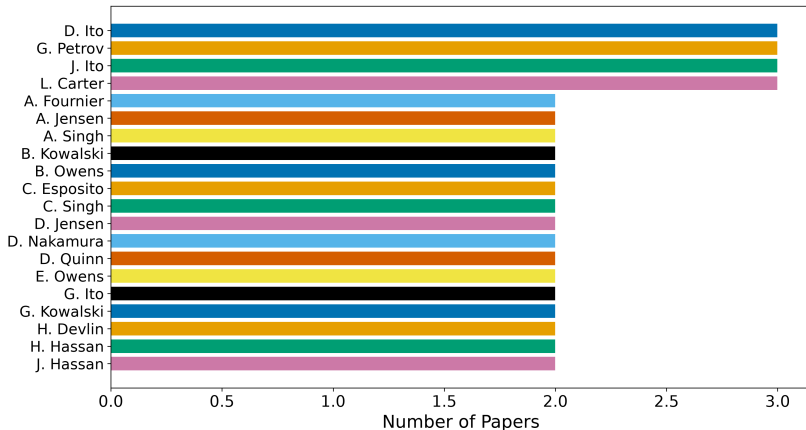


Figure 7: Author productivity for Exoplanet Atmospheres. The horizontal bar chart shows authors with the most retained records; names and counts depend on source coverage and deduplication.